

Karnaugh Map

The Karnaugh Map also called as K Map is a graphical representation that provides a systematic method for simplifying the Boolean expressions.

For a boolean expression consisting of n -variables, number of cells required in K Map = 2^n cells.

What is Karnaugh Map (K-Map)

A Karnaugh map (K-map) is a visual method used to simplify the algebraic expressions in Boolean functions without having to resort to complex theorems or equation manipulations. A K-map can be thought of as a special version of a truth table that makes it easier to map out parameter values and arrive at a simplified Boolean expression.

Two Variable K Map:

- Two variable K Map is drawn for a boolean expression consisting of two variables.
- The number of cells present in two variable K Map = $2^2 = 4$ cells.
- So, for a boolean function consisting of two variables, we draw a 2×2 K Map.

Two variable K Map may be represented as-

		B	
		0	1
A	0	$A'B'$ 0	$A'B$ 1
	1	AB' 2	AB 3

Here, A and B are the two variables of the given boolean function.

Three Variable K Map:

- Three variable K Map is drawn for a boolean expression consisting of three variables.
- The number of cells present in three variable K Map = $2^3 = 8$ cells.
- So, for a boolean function consisting of three variables, we draw a 2 x 4 K Map.

A	BC			
	$\overline{B} \overline{C}$	$\overline{B} C$	BC	$B \overline{C}$
$\overline{A}$	0	1	3	2
A	4	5	7	6

OR

A	BC			
	00	01	11	10
0	0	1	3	2
1	4	5	7	6

Here, A, B and C are the three variables of the given boolean function.

Four Variable K Map:

- Four variable K Map is drawn for a boolean expression consisting of four variables.
- The number of cells present in four variable K Map = $2^4 = 16$ cells.
- So, for a boolean function consisting of four variables, we draw a 4 x 4 K Map.

AB	CD			
	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	3	2
$\bar{A}B$	4	5	7	6
AB	12	13	15	14
$A\bar{B}$	8	9	11	10



AB	CD			
	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

Karnaugh Map Simplification Rules-

To minimize the given boolean function,

- We draw a K Map according to the number of variables it contains.
- We fill the K Map with 0's and 1's according to its function.
- Then, we minimize the function in accordance with the following rules.

Rule-01:

- We can either group 0's with 0's or 1's with 1's but we cannot group 0's and 1's together.
- X representing don't care can be grouped with 0's as well as 1's.

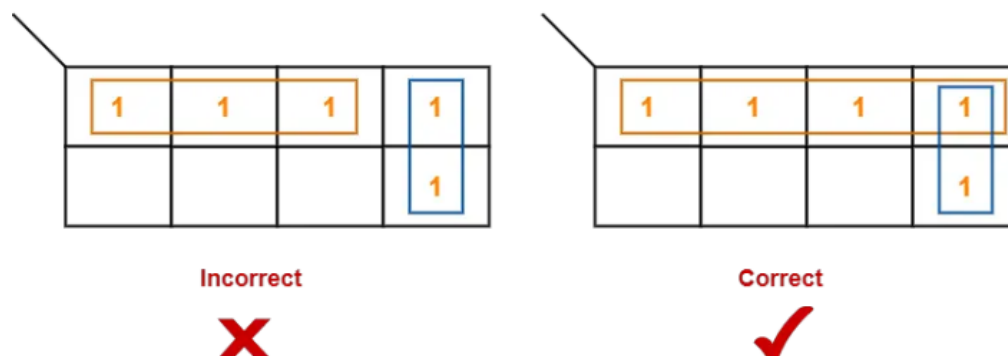
Rule-02:

- Groups may overlap each other.

Rule-03:

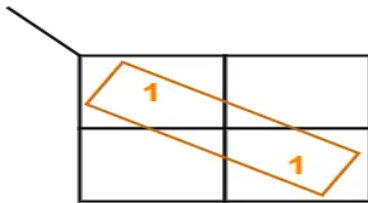
- We can only create a group whose number of cells can be represented in the power of 2.
- In other words, a group can only contain 2^n i.e. 1, 2, 4, 8, 16 and so on number of cells.

Example-

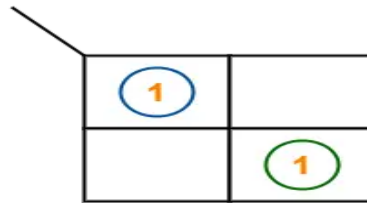


Rule-04:

- Groups can be only either horizontal or vertical.
- We cannot create groups of diagonal or any other shape.



Incorrect



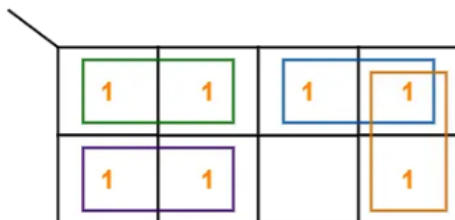
Correct



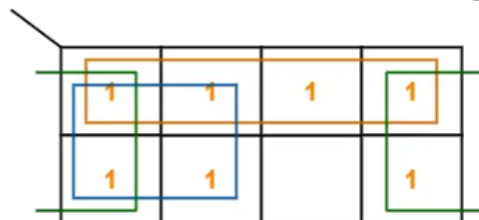
Rule-05:

- Each group should be as large as possible.

Example:



Incorrect



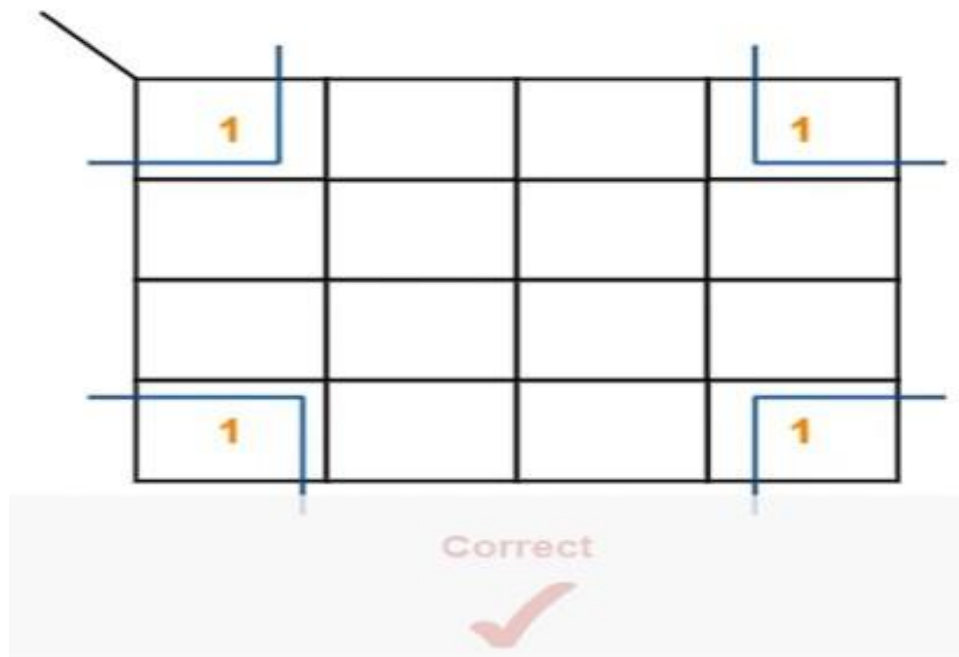
Correct



Rule-06:

- Opposite grouping and corner grouping are allowed.
- The example of opposite grouping is shown illustrated in Rule-05.

- The example of corner grouping is shown below.



Problem-01:

Minimize the following boolean function-

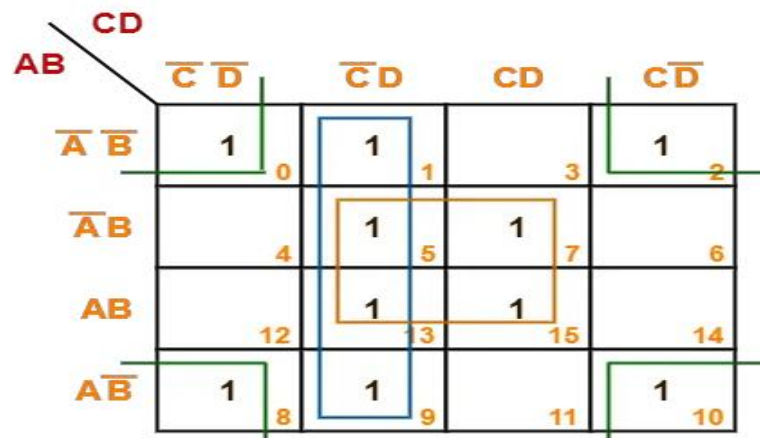
$$F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$$

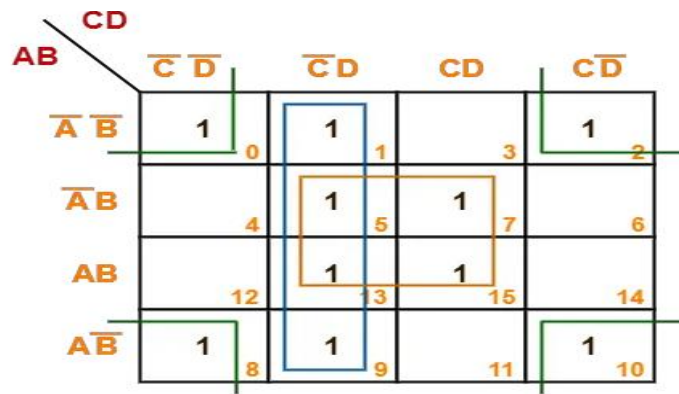
Solution:

- Since the given boolean expression has 4 variables, so we draw a 4 x 4 K Map.
- We fill the cells of K Map in accordance with the given boolean function.
- Then, we form the groups in accordance with the above rules.

Then, we

have-





Now,

$F(A, B, C, D)$

$$= (A'B + AB)(C'D + CD) + (A'B' + A'B + AB + AB')C'D + (A'B' + AB')(C'D' + CD')$$

$$= BD + C'D + B'D'$$

Problem-02:

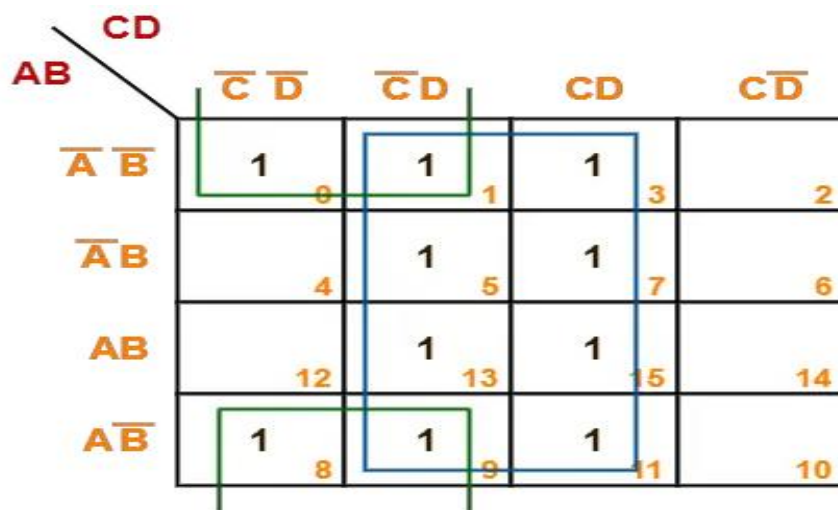
Minimize the following boolean function-

$$F(A, B, C, D) = \sum m(0, 1, 3, 5, 7, 8, 9, 11, 13, 15)$$

Solution-

- Since the given boolean expression has 4 variables, so we draw a 4 x 4 K Map.
- We fill the cells of K Map in accordance with the given boolean function.
- Then, we form the groups in accordance with the above rules.

Then, we have-



		CD			
		$\overline{C}\overline{D}$	$\overline{C}D$	CD	$C\overline{D}$
AB	$\overline{A}\overline{B}$	1	1	1	
	$\overline{A}B$		1	1	
	AB		1	1	
	$A\overline{B}$	1	1	1	

$$= (A'B' + A'B + AB + AB') (C'D + CD) + (A'B' + AB') (C'D' + C'D)$$

$$= D + B'C'$$

Thus, minimized boolean expression is:

$$F(A, B, C, D) = B'C' + D$$

Problem-03:

Minimize the following boolean function-

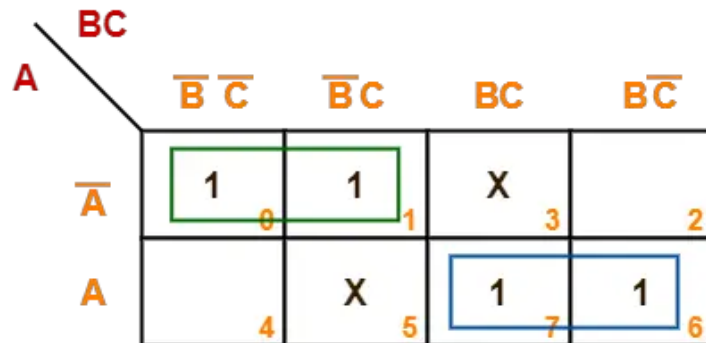
$$F(A, B, C) = \sum m (0, 1, 6, 7)$$

Solution-

- Since the given boolean expression has 3 variables, so we draw a 2 x 4 K Map.
- We fill the cells of K Map in accordance with the given boolean function.
- Then, we form the groups in accordance with the above rules.

Then, we have-

		BC			
		$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
A	$\overline{A}$	1	1	X	
	A		X	1	1



Now,

$F(A, B, C)$

$$= A'(B'C' + B'C) + A(BC + BC')$$

$$= A'B' + AB$$

Thus, minimized boolean expression is-

$$F(A, B, C) = AB + A'B'$$

K-map applications area:

K-Maps can be used to simplify these complex designs and reduce the number of logic gates required. K-Maps provide a graphical representation of the Boolean expressions, making it easier to visualize and understand the logic. K-Maps also provide a systematic method of simplifying complex expressions.

Purpose of the Karnaugh map:

Karnaugh maps are used to simplify real-world logic requirements so that they can be implemented using a minimum number of logic gates. A sum-of-products expression (SOP) can always be implemented using AND gates feeding into an OR gate, and a product-of-sums expression (POS) leads to OR gates feeding an AND gate.